

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**Product Identifier**

**Product Description:** Celite® 545  
**Cat No. :** C34967  
**CAS No** 68855-54-9  
**Molecular Formula** O2 Si

**Relevant identified uses of the substance or mixture and uses advised against**

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

**Details of the supplier of the safety data sheet**
**Importer**

Fisher Scientific Korea  
 D5,D6, Incheon Airport Logistics Complex  
 150, Gonghangdong-Ro 296 Beon-Gil  
 Jung-Gu, Incheon  
 Tel: +82-1661-9555  
 Fax: +82-2-2023-0603

**Supplier**

Thermo Fisher Scientific Chemicals, Inc.  
 30 Bond Street  
 Ward Hill, MA 01835-8099

**E-mail address** Chem.KR@thermofisher.com

**Emergency Telephone Number**

Emergency telephone: Medical: +(82) 070-7686-0086 or + 1-703-741-5970  
 CHEMTREC: 080 822 1374 (Local), CHEMTREC: 1-800-424-9300 or + 1-703-527-3887  
 Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

## SECTION 2: HAZARDS IDENTIFICATION

**Classification of the substance or mixture**
**Physical hazards**

Based on available data, the classification criteria are not met

**Health hazards**

Specific target organ toxicity - (repeated exposure)

Category 1

**Environmental hazards**

Based on available data, the classification criteria are not met

**Label Elements**

# SAFETY DATA SHEET

Celite® 545

Revision Date 12-Jun-2024



**Signal Word**

**Danger**

## **Hazard Statements**

H372 - Causes damage to organs through prolonged or repeated exposure

## **Precautionary Statements**

### **Prevention**

P260 - Do not breathe dust/fume/gas/mist/vapors/spray

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

### **Response**

P314 - Get medical advice/attention if you feel unwell

### **Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

## **Other Hazards**

This product does not contain any known or suspected endocrine disruptors

## **NFPA**

**Health**  
0

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### **3.1. Substances**

| Component                          | Common Name              | CAS No     | Index No | Weight % |
|------------------------------------|--------------------------|------------|----------|----------|
| Kieselguhr, soda ash flux-calcined | No information available | 68855-54-9 | KE-21796 | 99 - 100 |

## SECTION 4: FIRST AID MEASURES

### **Description of first aid measures**

#### **General Advice**

If symptoms persist, call a physician.

#### **Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### **Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

#### **Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

#### **Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

#### **Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to

# SAFETY DATA SHEET

Celite® 545

Revision Date 12-Jun-2024

protect themselves and prevent spread of contamination.

## **Most important symptoms and effects, both acute and delayed**

None reasonably foreseeable.

## **Indication of any immediate medical attention and special treatment needed**

### **Notes to Physician**

Treat symptomatically.

## **SECTION 5: FIREFIGHTING MEASURES**

### **Extinguishing media**

#### **Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### **Special hazards arising from the substance or mixture**

Thermal decomposition can lead to release of irritating gases and vapors.

### **Hazardous Combustion Products**

None under normal use conditions.

### **Advice for fire-fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### **Personal Precautions, Protective Equipment and Emergency Procedures**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

### **Environmental precautions**

Should not be released into the environment.

### **Methods and Material for Containment and Cleaning Up**

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

### **Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## **SECTION 7: HANDLING AND STORAGE**

### **Precautions for Safe Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

### **Conditions for Safe Storage, Including any Incompatibilities**

Keep container tightly closed in a dry and well-ventilated place.

# SAFETY DATA SHEET

Celite® 545

Revision Date 12-Jun-2024

## Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component                          | CAS No     | Korea      | ACGIH TLV  | OSHA PEL   |
|------------------------------------|------------|------------|------------|------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | Not listed | Not listed | Not listed |

| Component                          | CAS No     | European Union | The United Kingdom | Germany                                                                                   |
|------------------------------------|------------|----------------|--------------------|-------------------------------------------------------------------------------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | Not listed     | Not listed         | TWA: 0.3 mg/m <sup>3</sup> (8 Stunden). AGW - TWA: 0.3 mg/m <sup>3</sup> (8 Stunden). MAK |

### ACGIH - Biological Exposure Indices

| Component                          | CAS No     | ACGIH - Biological Exposure Indices |
|------------------------------------|------------|-------------------------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | Not listed                          |

### Exposure Controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### Personal protective equipment

##### Eye Protection

Wear safety glasses with side shields (or goggles)

##### Hand Protection

Protective gloves

##### Skin and body protection

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

##### Personal protective equipment

##### Respiratory Protection

Use only those certified by the Korea Occupational Safety and Health Administration.

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

##### Recommended Filter type:

Particulates filter conforming to EN 143

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

### Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

ALFAAC34967

# SAFETY DATA SHEET

Celite® 545

Revision Date 12-Jun-2024

## Information on basic physical and chemical properties

Appearance (Physical State, Color, etc.) Solid Powder

Odor No information available  
Odor Threshold No data available  
pH 10 10% aqueous solution

Melting Point/Range No data available  
Softening Point No data available  
Boiling Point/Range No information available  
Flash Point No information available **Method -** No information available

Evaporation Rate Not applicable Solid  
Flammability (solid,gas) No information available  
Explosion Limits No data available

Vapor Pressure No data available  
Vapor Density Not applicable Solid  
Specific Gravity / Density No data available  
Bulk Density No data available  
Water Solubility Insoluble  
Solubility in other solvents No information available

## Partition Coefficient (n-octanol/water)

| Component                          | CAS No     | log Pow           |
|------------------------------------|------------|-------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | No data available |

Autoignition Temperature No data available  
Decomposition Temperature No data available  
Viscosity Not applicable Solid  
Explosive Properties No information available  
Oxidizing Properties No information available

Molecular Formula O2 Si  
Molecular Weight 60.08

## SECTION 10: STABILITY AND REACTIVITY

Reactivity None known, based on information available

Chemical Stability Stable under normal conditions.

## Possibility of Hazardous Reactions

Hazardous Polymerization No information available.  
Hazardous Reactions None under normal processing.

Conditions to Avoid None known.

# SAFETY DATA SHEET

Celite® 545

Revision Date 12-Jun-2024

## Incompatible Materials

None known.

## Hazardous Decomposition Products

None under normal use conditions.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### Information on expected route of exposure

**Inhalation** Avoid breathing dust or spray mist. May be harmful if inhaled.  
**Ingestion** May be harmful if swallowed.  
**Eyes** Avoid contact with eyes.  
**Skin** Avoid contact with skin.

#### Information on Health Hazards

##### (a) acute toxicity;

**Oral** No data available  
**Dermal** No data available  
**Inhalation** No data available

| Component                          | CAS No     | LD50 Oral         | LD50 Dermal       | LC50 Inhalation                |
|------------------------------------|------------|-------------------|-------------------|--------------------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | No data available | No data available | LC50 > 2.6 mg/L ( Rat )<br>4 h |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

##### (d) respiratory or skin sensitization;

**Respiratory** No data available  
**Skin** No data available

| Component                          | CAS No     | Test method       | Test species      | Study result      |
|------------------------------------|------------|-------------------|-------------------|-------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | No data available | No data available | No data available |

(e) germ cell mutagenicity; No data available

| Component                          | CAS No     | Test method       | Test species      | Study result      |
|------------------------------------|------------|-------------------|-------------------|-------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | No data available | No data available | No data available |

(f) carcinogenicity; No data available

| Component                          | CAS No     | Test method       | Test species / Duration | Study result      |
|------------------------------------|------------|-------------------|-------------------------|-------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | No data available | No data available       | No data available |

There are no known carcinogenic chemicals in this product

| Component            | CAS No     | IARC       | NTP        | ACGIH      | OSHA       | UK         |
|----------------------|------------|------------|------------|------------|------------|------------|
| Kieselguhr, soda ash | 68855-54-9 | Not listed | Not listed | Not listed | Not listed | Not listed |

# SAFETY DATA SHEET

Celite® 545

Revision Date 12-Jun-2024

|               |  |  |  |  |  |  |
|---------------|--|--|--|--|--|--|
| flux-calcined |  |  |  |  |  |  |
|---------------|--|--|--|--|--|--|

(g) reproductive toxicity; No data available

| Component                          | CAS No     | Test method       | Test species / Duration | Study result      |
|------------------------------------|------------|-------------------|-------------------------|-------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | No data available | No data available       | No data available |

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; Category 1

Target Organs No information available.

(j) aspiration hazard; Not applicable  
Solid

Other Adverse Effects  
No information available.

| Component                          | CAS No     | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances | Japan - Endocrine Disruptor Information |
|------------------------------------|------------|------------------------------------------|--------------------------------------------------|-----------------------------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | Not applicable                           | Not applicable                                   | Not applicable                          |

## SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

| Component                          | CAS No     | Freshwater Fish   | Water Flea        | Freshwater Algae  | Microtox          |
|------------------------------------|------------|-------------------|-------------------|-------------------|-------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | No data available | No data available | No data available | No data available |

Persistence and degradability

Persistence Insoluble in water.  
Degradability Not relevant for inorganic substances.

Bioaccumulative potential May have some potential to bioaccumulate

Mobility in soil Spillage unlikely to penetrate soil Is not likely mobile in the environment due its low water solubility.

Ozone Depletion Potential

| Component                          | CAS No     | Ozone Depletion Potential |
|------------------------------------|------------|---------------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | Not listed                |

Other adverse effects No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods

Waste from Residues/Unused Products Waste is classified as hazardous. Dispose in accordance with the Wastes Control Act (폐기물관리법).

# SAFETY DATA SHEET

Celite® 545

Revision Date 12-Jun-2024

|                               |                                                                                                                               |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Contaminated Packaging</b> | Dispose of this container to hazardous or special waste collection point.                                                     |
| <b>Other Information</b>      | Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. |

## SECTION 14: TRANSPORT INFORMATION

|                                |                       |
|--------------------------------|-----------------------|
| <b>Road and Rail Transport</b> | Not Regulated         |
| <b>IATA</b>                    | Not regulated         |
| <b>IMDG/IMO</b>                | Not regulated         |
| <b>Marine Pollutant</b>        | No hazards identified |

|                                     |                                 |
|-------------------------------------|---------------------------------|
| <b>Special Precautions for User</b> | No special precautions required |
|-------------------------------------|---------------------------------|

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

**Legend:** X - Listed '-' - Not Listed

#### International Inventories

| Component                          | CAS No     | KECL     | TSCA | EINECS    | IECSC | DSL | NDSL | PICCS | ENCS | ISHL | AICS |
|------------------------------------|------------|----------|------|-----------|-------|-----|------|-------|------|------|------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | KE-21796 | X    | 272-489-0 | X     | X   | -    | X     | X    | X    | X    |

| Component                          | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

| Component                          | CAS No     | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential |
|------------------------------------|------------|----------|------------------------------|---------------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | Listed   | Not applicable               | Not applicable            |

#### Korean National Regulations

| Component                          | CAS No     | Act on Registration and Evaluation of Chemical Substances (K-REACH) | Authorised Chemicals | Existing Substances Subject to Registration |
|------------------------------------|------------|---------------------------------------------------------------------|----------------------|---------------------------------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | Annex 1 - KE-21796                                                  | Not applicable       | Not applicable                              |

| Component                          | CAS No     | Chemical Control Act - Toxic Chemicals | Chemical Control Act - Prohibited Chemicals | Chemical Control Act - Use Restricted Chemicals |
|------------------------------------|------------|----------------------------------------|---------------------------------------------|-------------------------------------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | Not applicable                         | Not applicable                              | Not applicable                                  |

| Component                          | CAS No     | Chemical Control Act - Accident Precaution Chemicals (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures) |
|------------------------------------|------------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | Not applicable                                                       | Not applicable                                                                                 | Not applicable                                                                                         |

| Component | CAS No | Waste Control Law | Ministry of Environment | Ministry of Environment |
|-----------|--------|-------------------|-------------------------|-------------------------|
|-----------|--------|-------------------|-------------------------|-------------------------|

ALFAAC34967



# SAFETY DATA SHEET

Celite® 545

Revision Date 12-Jun-2024

|                                    |            |                | CMR risk       | Critically Controlled Substance |
|------------------------------------|------------|----------------|----------------|---------------------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | Not applicable | Not applicable | Not applicable                  |

| Component                          | CAS No     | ISHA - Harmful Agents Subject to Work Environment Monitoring | ISHA - Prohibited substances | ISHA - Substances requiring permission |
|------------------------------------|------------|--------------------------------------------------------------|------------------------------|----------------------------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | Not applicable                                               | Not applicable               | Not applicable                         |

| Component                          | CAS No     | ISHA - Substances subject to control | ISHA - Harmful Agents Requiring Health Examination | ISHA - Permissible Exposure Limits |
|------------------------------------|------------|--------------------------------------|----------------------------------------------------|------------------------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | Not applicable                       | Not applicable                                     | Not applicable                     |

| Component                          | CAS No     | ISHA - Subject to Process Safety Reports (minimum quantity) | ISHA - Threshold Limit Values (TLVs) Chemicals | ISHA - Special management materials |
|------------------------------------|------------|-------------------------------------------------------------|------------------------------------------------|-------------------------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | Not applicable                                              | Not applicable                                 | Not applicable                      |

**National Fire Association - Dangerous Substances** Minimum quantity requiring a permit

| Component                          | CAS No     | Class 1 - Oxidising solids | Class 2 - Flammable solid | Class 3 - Spontaneously Combustible Substances and Dangerous Substances When Wet | Class 4 - Flammable liquids | Class 5 - Self-reactive substances | Class 6 - Oxidising liquids |
|------------------------------------|------------|----------------------------|---------------------------|----------------------------------------------------------------------------------|-----------------------------|------------------------------------|-----------------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | Not applicable             | Not applicable            | Not applicable                                                                   | Not applicable              | Not applicable                     | Not applicable              |

## Control Parameters

| Component                          | CAS No     | Korea      | ACGIH - Biological Exposure Indices |
|------------------------------------|------------|------------|-------------------------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | Not listed | Not listed                          |

## US Management Information

OSHA - Occupational Safety and Health Administration

Not applicable

| Component                          | CAS No     | Specifically Regulated Chemicals | Highly Hazardous Chemicals |
|------------------------------------|------------|----------------------------------|----------------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | Not applicable                   | Not applicable             |

**CERCLA** Not applicable

| Component                          | CAS No     | CERCLA Extremely Hazardous Substances RQs | Hazardous Substances RQs | SARA 313 - Threshold Values % |
|------------------------------------|------------|-------------------------------------------|--------------------------|-------------------------------|
| Kieselguhr, soda ash flux-calcined | 68855-54-9 | Not applicable                            | Not applicable           | Not applicable                |

## GHS Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

Danger.

H372 - Causes damage to organs through prolonged or repeated exposure.

P260 - Do not breathe dust/fume/gas/mist/vapors/spray. P314 - Get medical advice/attention if you feel unwell.

## SECTION 16: OTHER INFORMATION

# SAFETY DATA SHEET

Celite® 545

Revision Date 12-Jun-2024

## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

## Prepared By

Health, Safety and Environmental Department

## Creation Date

30-Jan-2023

## Revision Date

12-Jun-2024

## Revision Number

2

## Revision Summary

New emergency telephone response service provider.

## MOEL's Public Notice No. 2023-9 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**